

EdiGlobe VLSI Internship

Minor Project

Project Title: Realize JKFF using DFF and write the verilog code for a JK Flip flop, and its testbench. Use all possible combinations of inputs to test its working

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Software used: EDA playground

1. Objective

To design and implement a JK Flip-Flop using a D Flip-Flop in Verilog HDL and verify its functionality by simulating all possible input combinations using a testbench in EDA Playground.

2. Theory

A JK Flip-Flop is a sequential logic circuit that removes the invalid state of the SR Flip-Flop. A JK Flip-Flop can be implemented using a D Flip-Flop by generating the D input from the characteristic equation.

3. Characteristic Equation

$$D = J\bar{Q} + \bar{K}Q$$

4. Truth Table

J K	Operation	Next State
00	Hold	Q
01	Reset	0
10	Set	1
11	Toggle	$\bar{Q}$

5. Verilog Design Code

```
`timescale 1ns/1ps

module d_flipflop(
    input d,
    input clk,
    output reg q
);
always @(posedge clk)
begin
    q <= d;
end
endmodule

module jk_flipflop(
    input j,
    input k,
    input clk,
    output q
);
wire d;
wire q_internal;
assign d = (j & ~q_internal) | (~k & q_internal);
d_flipflop ff1(
    .d(d),
    .clk(clk),
```

```
        .q(q_internal)
    );
    assign q = q_internal;
endmodule
```

6. Testbench Code

```
`timescale 1ns/1ps
module testbench;
    reg j;
    reg k;
    reg clk;
    wire q;
    jk_flipflop uut(
        .j(j),
        .k(k),
        .clk(clk),
        .q(q)
    );
    always #5 clk = ~clk;
    initial
    begin
        $dumpfile("jk_flipflop.vcd");
        $dumpvars(0, testbench);
    end
endmodule
```

```
    clk = 0;

    $display("%-12s %-6s %-6s %-6s", "Time", "J", "K", "Q");
$monitor("%-12d %-6b %-6b %-6b",

    $time,

    j,

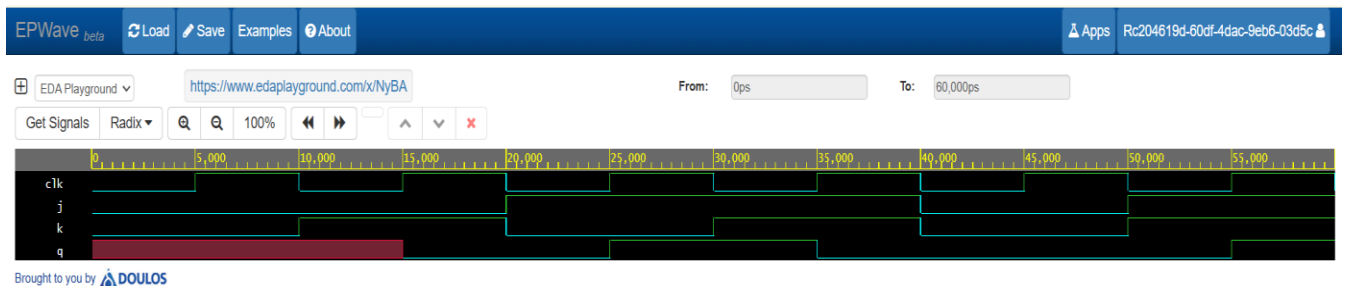
    k,

    q);
j = 0; k = 0;
#10;
j = 0; k = 1;
#10;
j = 1; k = 0;
#10;
j = 1; k = 1;
#10;
j = 0; k = 0;
#10;
j = 1; k = 1;
#10;
$finish;
end
endmodule
```

7. Simulation Output

```
Log Share
[2026-07-14 07:18:32 UTC] iverilog '-wall' '-g2012' design.sv testbench.sv && unbuffer vvp a.out
VCD info: dumpfile jk_flipflop.vcd opened for output.
Time      J      K      Q
0         0      0      x
10        0      1      x
15        0      1      0
20        1      0      0
25        1      0      1
30        1      1      1
35        1      1      0
40        0      0      0
50        1      1      0
55        1      1      1
testbench.sv:52: $finish called at 60000 (1ps)
Finding VCD file...
./jk_flipflop.vcd
[2026-07-14 07:18:33 UTC] Opening EPWave...
Done
```

8. Waveform



9. Result

The JK Flip-Flop was successfully realized using a D Flip-Flop. Simulation verified the Hold, Reset, Set and Toggle operations for all possible input combinations.

10. Conclusion

The project demonstrates the implementation and verification of a JK Flip-Flop using Verilog HDL. The design compiled successfully and the simulation results matched the expected behavior.